VLSI PROJEKAT - IZVEŠTAJ

Projektovanje VLSI sistema (19E043VLSI)

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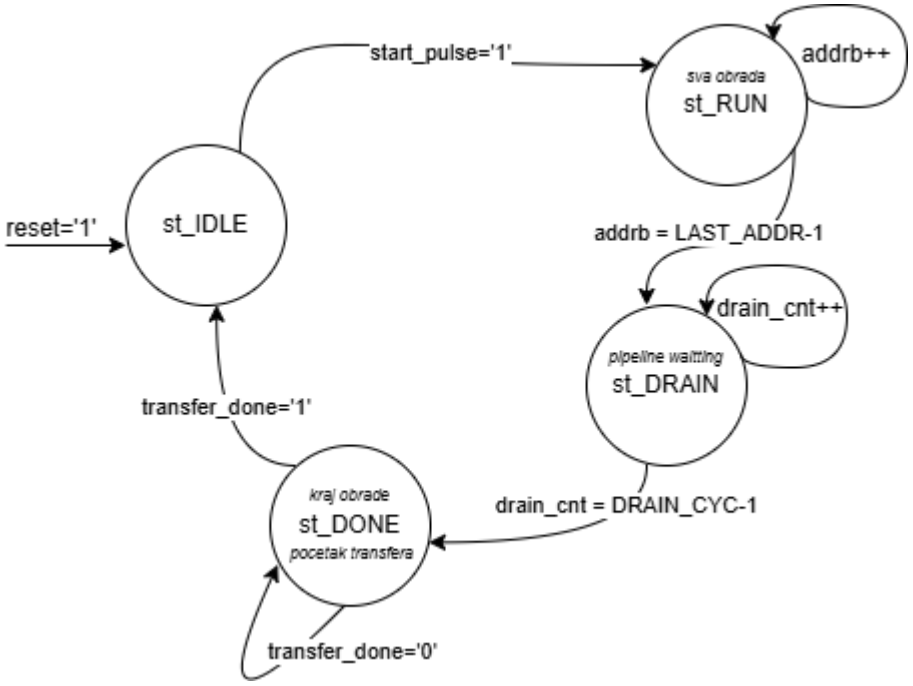
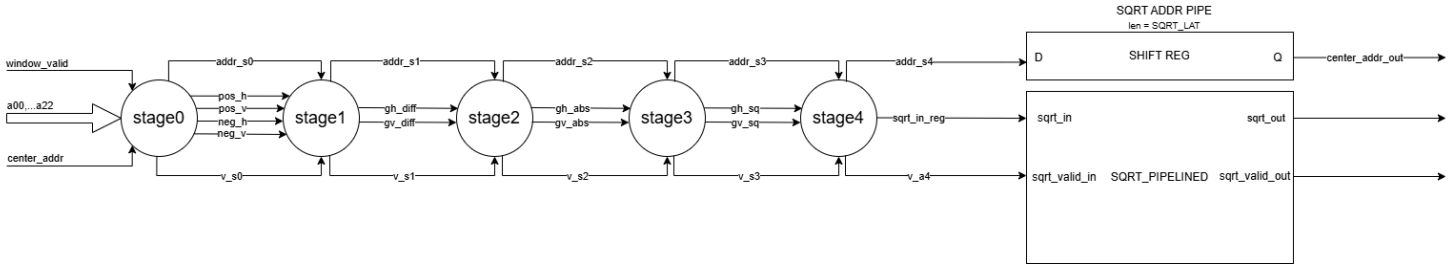
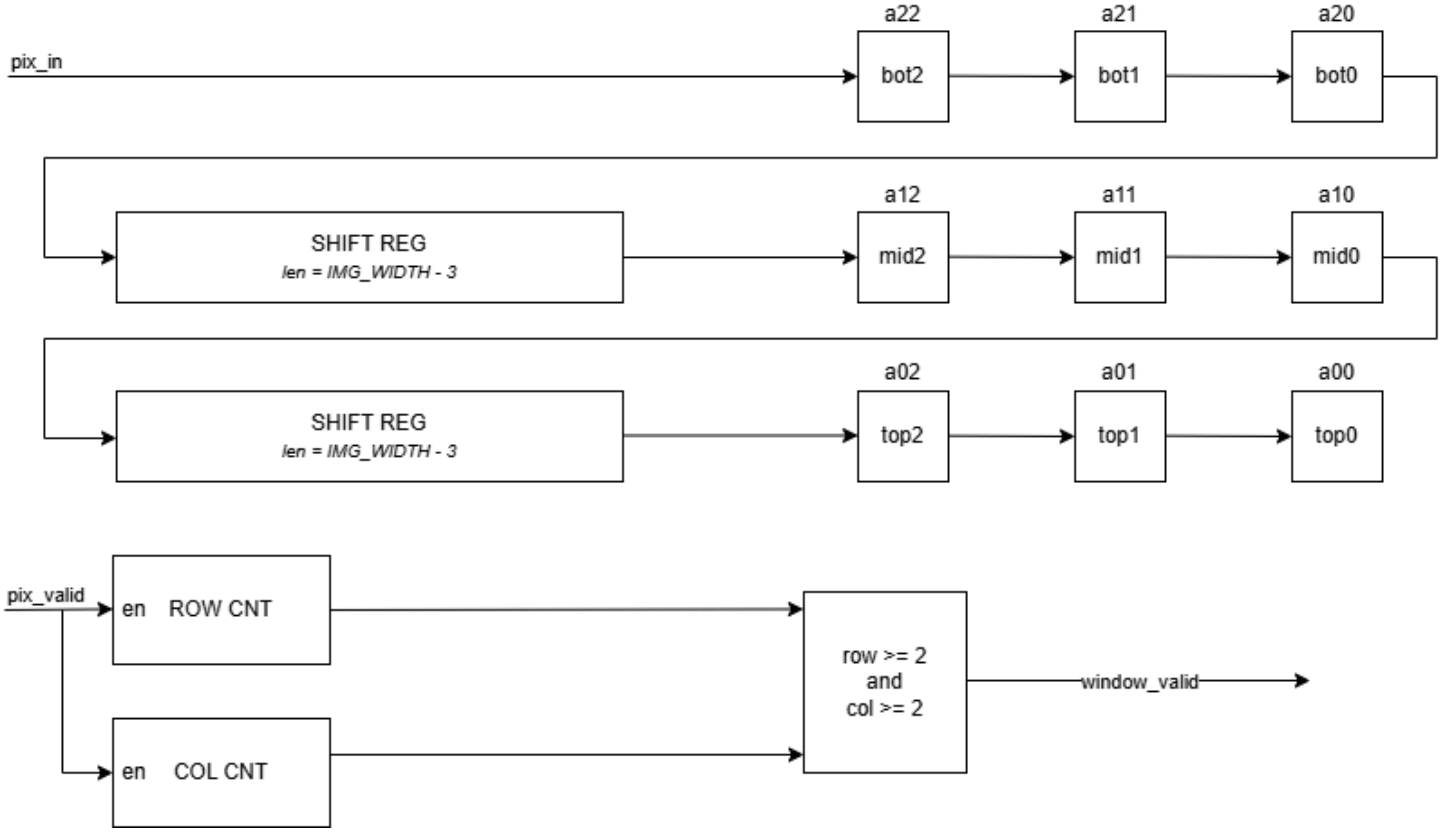
U nastavku će biti priložene blok šeme i mašine stanja ključnih modula:



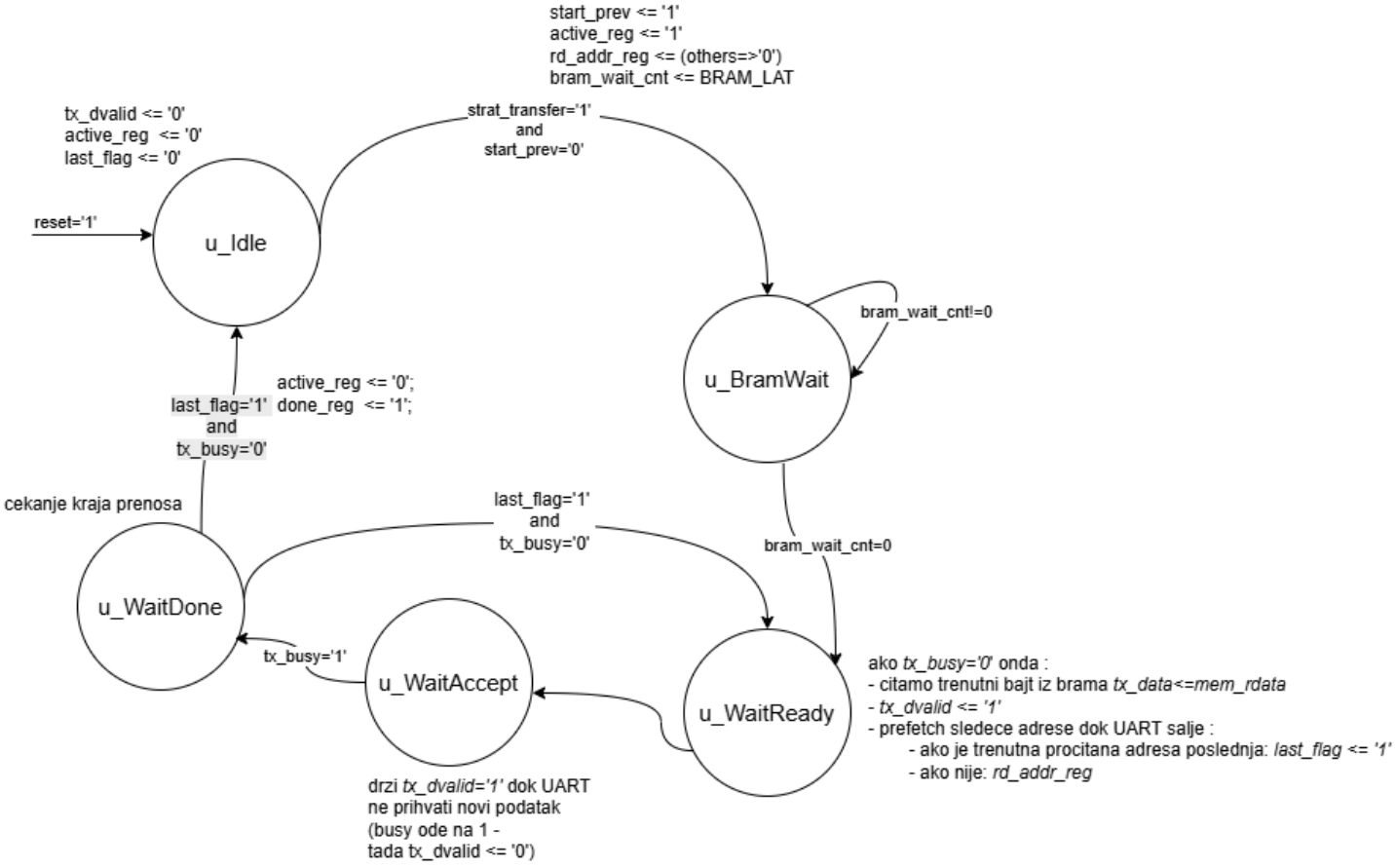
Mašina stanja celog sistema



Blok šema modula za primenu filtriranja slike *Sobelovim* filtrom (unutar ovog modula instanciran pajplanovani moduo za računanje korena, *sqrt_pipelined*) - *sobel_mag_sqrt_compute.vhd*

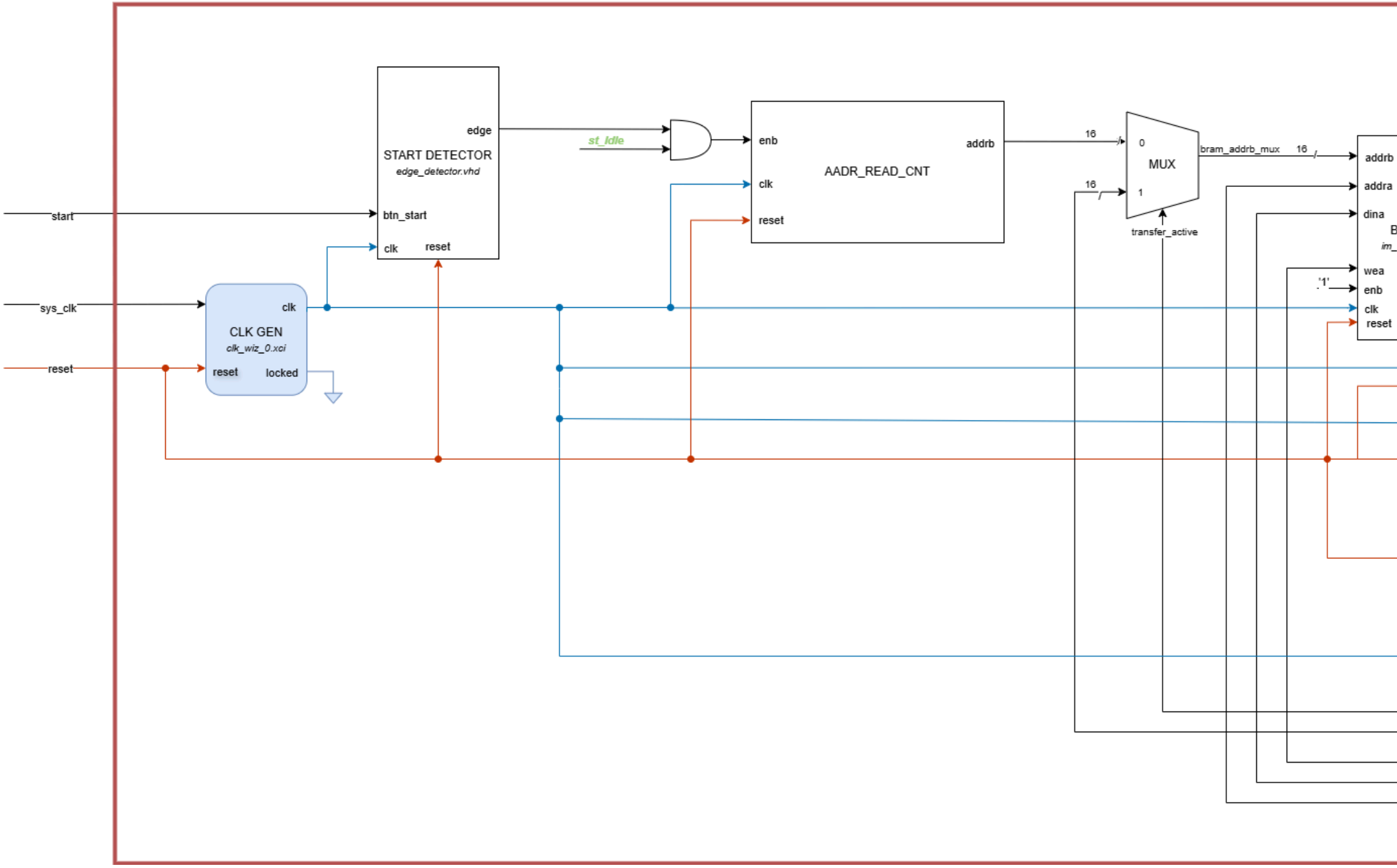


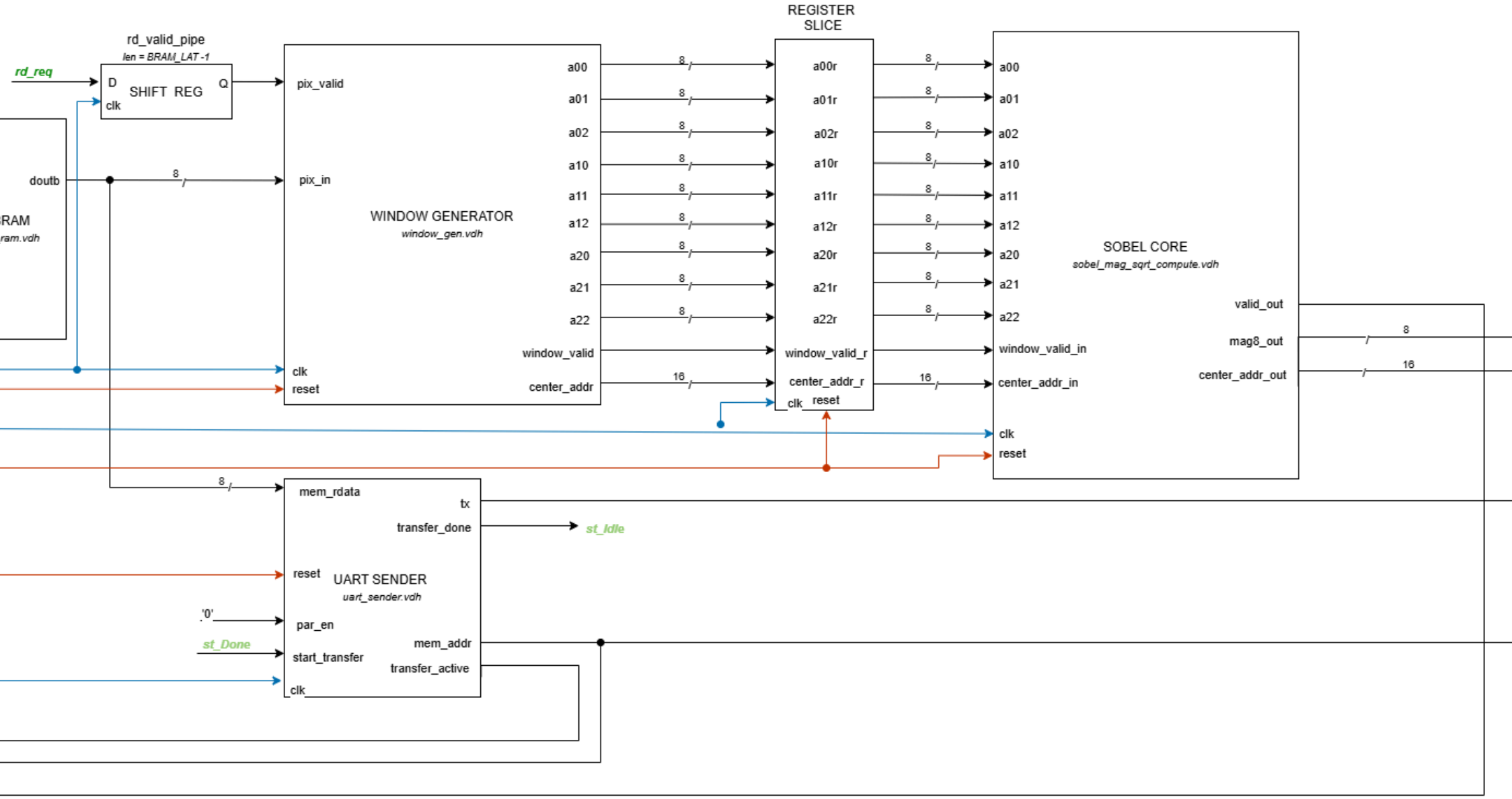
Blok šema modula za izdvajanje prozora validnih piksela za primenu Sobelovih operatora - *window_gen.vhd*

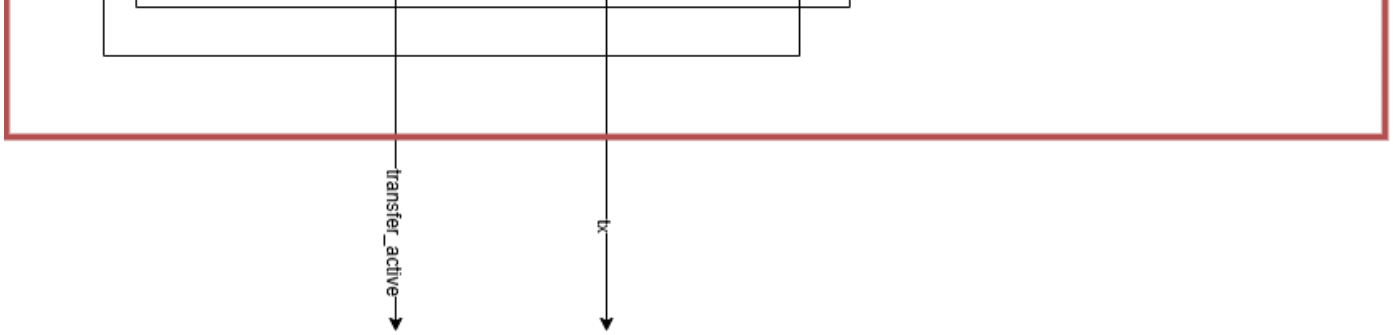


Mašina stanja modula uart_sender.vhd, odnosno kontrolera za asinhronu serijsku komunikaciju (uart) sa PC-jem, kojim se nakon završetka obrade šalje rezultat na PC

PROJEKAT TOP




rd_valid_pipe
len = BRAM_LAT -1
rd_req
D
SHIFT REG
Q
clk
doutb
RAM
ram.vdh
8
pix_valid
pix_in
WINDOW GENERATOR
window_gen.vdh
clk
reset
a00
a01
a02
a10
a11
a12
a20
a21
a22
window_valid
center_addr
16
REGISTER
SLICE
a00r
a01r
a02r
a10r
a11r
a12r
a20r
a21r
a22r
window_valid_r
center_addr_r
clk
reset
SOBEL CORE
sobel_mag_sqrt_compute.vdh
window_valid_in
center_addr_in
valid_out
mag8_out
center_addr_out
mem_rdata
tx
transfer_done
st_Idle
reset
UART SENDER
uart_sender.vdh
'0'
par_en
st_Done
start_transfer
mem_addr
transfer_active
clk



Šema celog sistema - idejno

Timing report i resursi za moduo za računanje korena (pajplajnovani) - nakon implementacije

LUT	FF	BRAM	DSP	*f_CLK* [MHz]	Kašnjenje/Latency - kritična putanja [ns]	Logic Delay - kritična putanja	Net Delay - kritična putanja	WNS	WHS
291	409	0	0	200	3.317	2.209	1.108	0.802	0.148

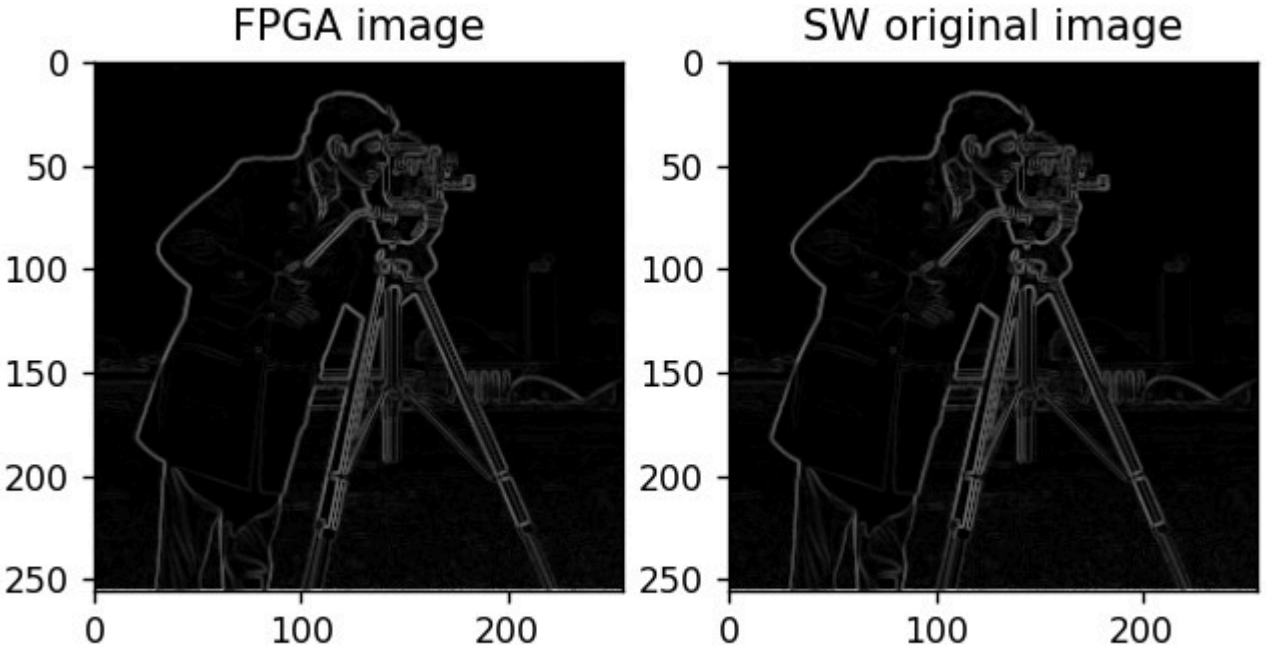
Maksimalna učestanost za ovu arhitekturu je onda ~235MHz

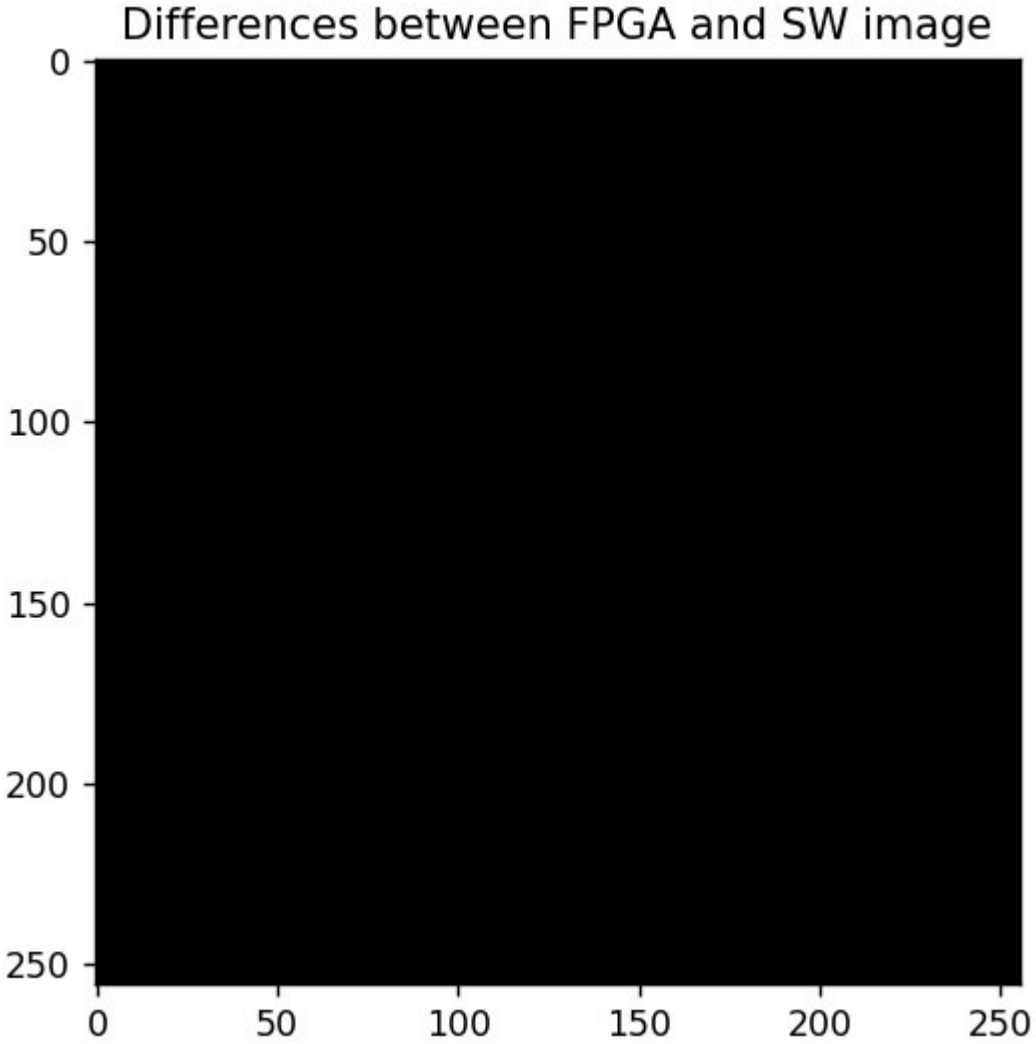
Timing report i resursi za ceo sistem - nakon implementacije

LUT	FF	BRAM	DSP	*f_CLK* [MHz]	Kašnjenje/Latency - kritična putanja [ns]	LogicDelay - kritična putanja	NetDelay - kritična putanja	WNS	WHS
507	1028	16	2	200	3.317	0.612	3.319	0.445	0.045

Maksimalna učestanost za ovu arhitekturu je onda ~219MHz

FPGA vs Python


FPGA image
SW original image
0
50
100
150
200
250
0
100
200


Differences between FPGA and SW image
0
50
100
150
200
250
0
50
100
150
200
250